

Concept abstract improved

(Background) Climate projections effectively estimate global surface temperature trends over a century which is mainly dominated by climate change. In the North Atlantic climate change impact on sea surface temperature (SST) has been comparatively modest and trends diverge significantly from the global trend. However, this region plays a crucial role in shaping the European climate making decadal SST trend predictions interesting. To predict SST trends, the analogue method searches for analogues of observed SST patterns within a simulation ensemble. Due to ocean inertia, initially similar SST patterns remain close for some time in the future, enabling the analogue method to predict SST evolution.

(Research Gap) While the analogue method has been used for decadal predictions, its application to the prediction of ten-year SST trends remains unexplored.

(Message) This thesis aims to extend the analogue method to predict ten-year trends using the MPI-ESM simulation set of 30 members.

(Methods) We reconstruct the well-observed time series from 1960 to 2020 using the analogue method and derive ten-year trends based on linear fits. To apply the analogue method for trend predictions, we analyze its dependence on various parameters used in previous works. As the method cannot time external forcings like volcanoes, we assess the influence of volcanic eruptions on trend skill.

(Results) We report that for one simulation ensemble, the MPI-ESM Grand Ensemble, the analogue method is robust. Trend time series reconstruction exhibits a correlation with the reference of up to 0.9, revealing an anticipated warming of the North Atlantic sea surface until 2029.

(Conclusion) The results suggest the effectiveness of the analogue method with a single simulation ensemble. A comparison of forecast skill shows that there is not one best parameter set.



Outline anhand von Stellvertretersätzen

1. Introduction Chapter Outline

General topic

Start with introductory sentence about trend (E.g.: Climate projections effectively estimate global surface temperature trends over a century, but historical North Atlantic sea surface temperature (SST) trends diverge significantly from the global trend.)

- Significance of trends for the knowledge about climate change impacts

Predictions of future climate evolution takes the form of projections of the next century, using different Shared Socioeconomic Pathways from the 6th Coupled Model Intercomparison Project (CMIP6), or initialized predictions of the next decade.

While projections are a good estimation of long-term climate change, they are imprecise on shorter time scales and for regional changes. The dominant source of uncertainty in the first decade is internal variability of the climate system, even before model and scenario uncertainties.

The idea of the analogue method is to phase in internal variability by comparing observed and simulated climate states and finding the most similar match to the observed climate state in the simulations.

Processes in the North Atlantic, such as the Atlantic subpolar gyre (SPG) and the Atlantic Meridional Overturning Circulation (AMOC), impact the European climate.

Oceans have a large heat capacity and inertia, meaning that changes in temperature happen slower than over land masses.

Knowns

To predict SST trends, the analogue method searches for analogues of observed SST patterns within a simulation ensemble.

Because of ocean inertia, two SST patterns that are initially similar will remain close for some time in the future, making it possible to predict future evolution of SST with the analogue method.

The analogue method is used in climate sciences and meteorology for simple prediction schemes. It can be performed using information from long historical records or simulation data sets, assuming the models providing these simulations are an adequate representation of reality.

Earth system models (ESM) are designed to simulate the most important features in the global climate, but are less precise on regional scales.

By selecting those simulation ensemble members that closely resemble the observed patterns of SST variability, the uncertainty arising from internal climate variability can be reduced.

Unknowns (Research Gap)

The analogue method has been used for decadal predictions of SST but not for the prediction of ten-year SST trends.

Menary *et al.* and Mahmood *et al.* used large archives of simulation data for the search of analogues, underlying the assumptions that better analogues can be found in larger data sets.

The sensitivity of the analogue method's ability to predict trends to the assumptions made about the main parameters is unclear.

Research topic

This thesis aims to extend the analogue method to predict ten-year trends using the MPI-ESM simulation set of 30 members.

As the method cannot time external forcings like volcanoes, we assess the influence of volcanic eruptions on trend skill.

In addition to time series reconstruction, we can make trend predictions until 2029.

Experimental approach

We reconstruct the well-observed time series from 1960 to 2020 using the analogue method and derive ten-year trends based on linear fits.

To apply the analogue method for trend predictions, we analyze its dependence on various parameters used in previous works.

For this purpose, we compare trend skill under the influence of volcanic upper ocean cooling with the skill under exclusion of volcano years.

3. Results Chapter Outline

Result 1: Trend time series reconstruction with the analogue method

(Goal: first in intro) This thesis aims to extend the analogue method to predict ten-year trends using the MPI-ESM simulation set of 30 members.

(Sub-Goal) We want to compare the ability of the analogue method to predict the major SST patterns in the North Atlantic, created by the North Atlantic subpolar gyre (SPG).

(Result) Reconstruction of the trend time series was successful and we find that the analogue methods has improved skill in these regions compared to the persistence/reconstruction made with the reference.

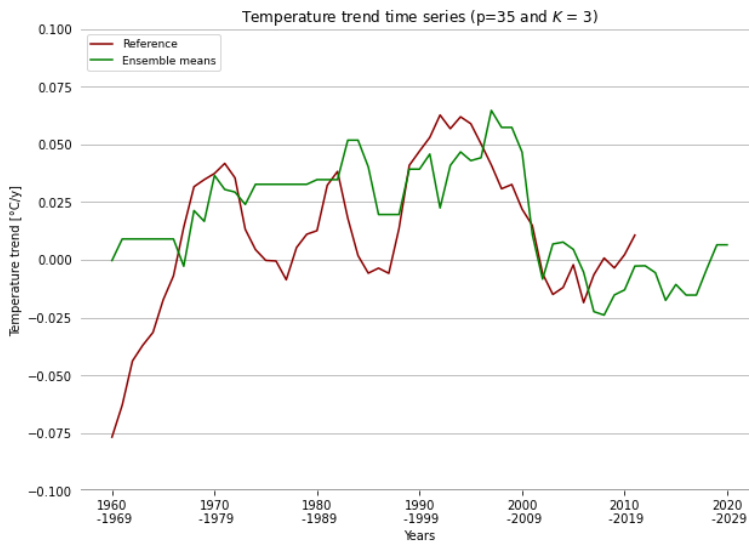


Fig. 4: The trend time series of the reference and one example analogue reconstruction done with the parameters and similarity metric suggested Menary, 2021. In three years (1963, 1982, 1991) a major volcanic eruption influenced the North Atlantic. These external events cannot be predicted by the analogue method (black circles). The correlation between reference and reconstruction with exclusion of volcanic years is $r = 0.575$

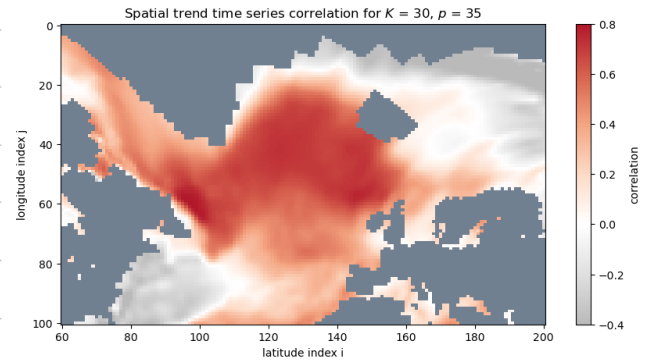


Fig. 5: Spatial trend skill.

Result ? : Volcanic influence

(Goal: *third in intro*) As the method cannot time external forcings like volcanoes, we assess the influence of volcanic eruptions on trend skill.

(Result) The temperature trend time series of the parameter for p used in Menary has been constructed. We can clearly see the influence of the volcanic eruptions. Skill of the reconstruction varies from 0.575 with the inclusion of these years to 0.788 without them. This influence can be seen for all parameter sets. Expecting the analogue method to predict external forcings like volcanoes, which it cannot do by design, significantly influences the skill.

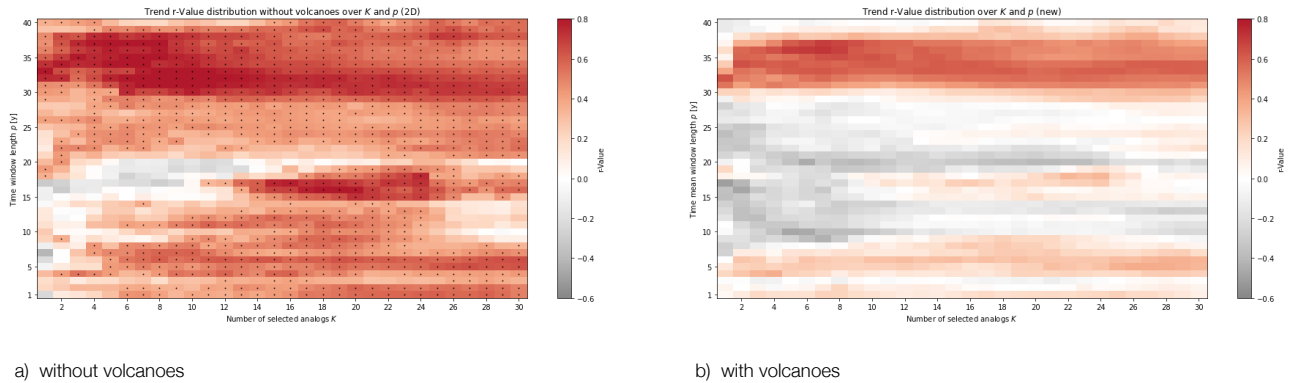


Fig. 6: Comparing trend skill for 1200 different parameter sets shows that expecting the method to predict volcano influence significantly alters the trend skill.

Result 2+3: Comparison of parameters and sensitivity

(Goal: second in intro) To apply the analogue method for trend predictions, we analyze its dependence on various parameters used in previous works. Similarity definitions were compared and a new definition of similarity that depends on the time evolution of the climate states was tested.

(Results) We report that for one simulation ensemble, the MPI-ESM Grand Ensemble, the analogue method is robust. The trend time series reconstruction exhibits a correlation with the reference of up to 0.9.

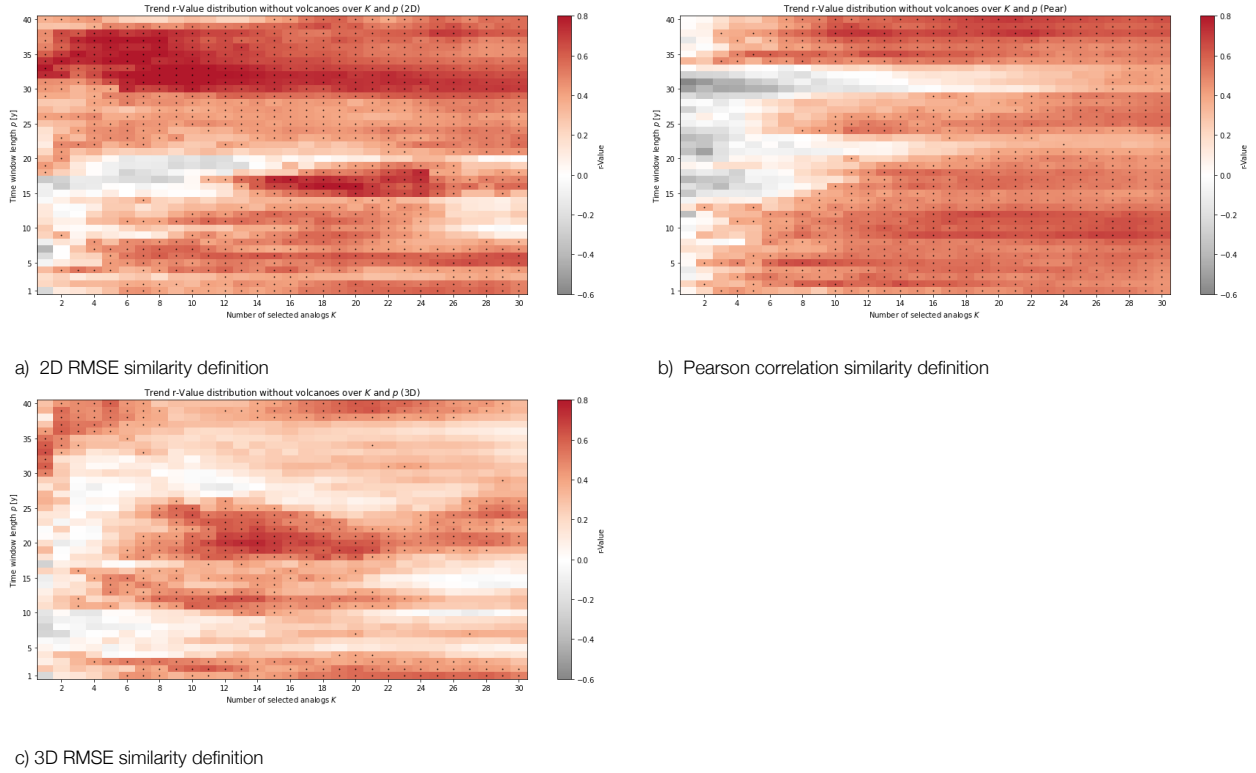


Fig. 7: Trend correlations for the systematic parameter analysis for p and K . a)-c) show different definitions of similarity. The correlations lie between 0.91 and -0.61, with a mean of 0.454. Stipples indicate values above significance of the 1000-member bootstrapped probability distribution at the 95% level. The mean 95% level lies at 0.36. a) shows the 2D RMSE definition, with $r_{\text{max}} = 0.91$, $r_{\text{min}} = -0.41$ and $r_{\text{mean}} = 0.436$. b) shows the Pearson correlation based definition, with $r_{\text{max}} = 0.72$, $r_{\text{min}} = -0.61$ and $r_{\text{mean}} = 0.37$. c) shows the 3D RMSE definition with $r_{\text{max}} = 0.76$, $r_{\text{min}} = -0.27$ and $r_{\text{mean}} = 0.29$.

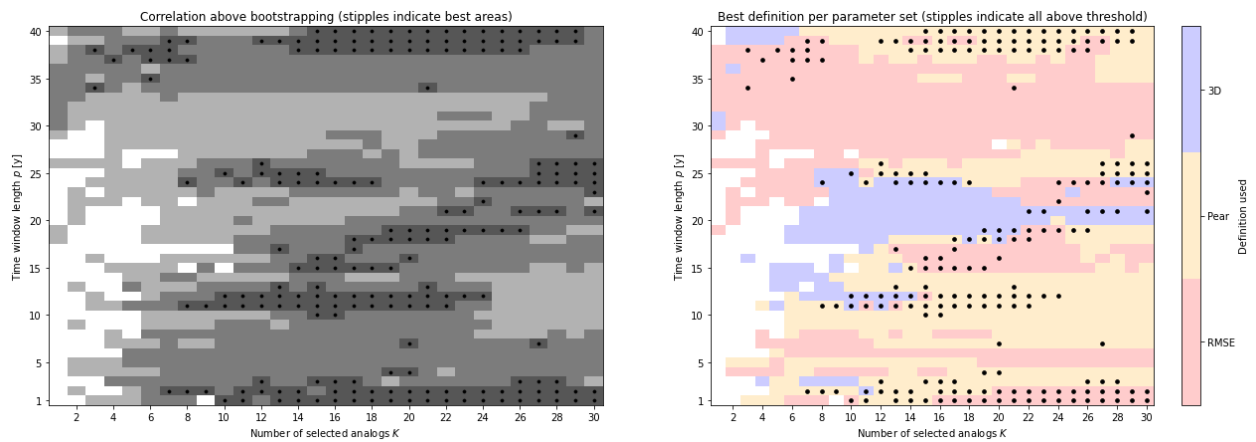


Fig. 8: Overview of parameter areas where one (light grey) or more (overlaps) similarity definitions exceed their trend correlation significance threshold. Stipples indicate where all three definitions are above their threshold.

Fig. 9: Overview of parameter sets meeting the significance threshold at the 95% level. Stipples indicate where all three similarity definitions are above their threshold level.

Result 3.2: Trend comparison

(Goal) Analysis of sensitivity to parameters for trend predictions with the analogue method.

(Result) The parameters used in previous works do not give best results for the application of the analogue method to trend predictions. Yet, there is not one best set of parameters.

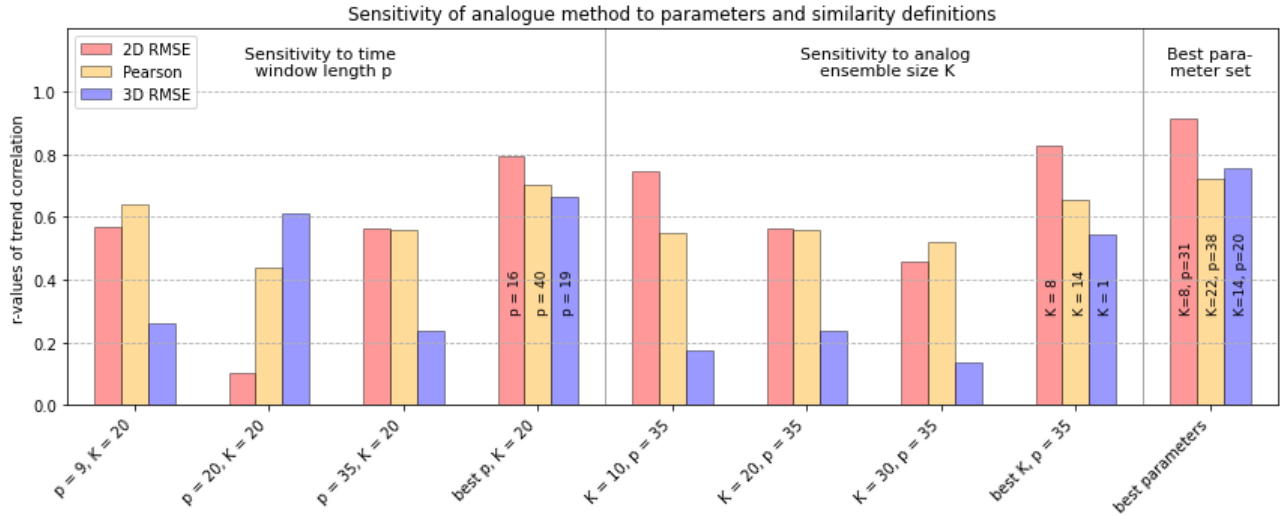


Fig. 10: Correlation of reconstructed trend time series. Comparison of similarity definition (colors) for different parameter sets. Sensitive to time window parameters from [Menary, Mahmood] with the best parameters found (left); the sensitivity of analog ensemble size (middle) and the best correlation values found for all parameters.

Result 4: Trend reconstruction and prediction until 2029

(Goal: last in intro) In addition to time series reconstruction, we make trend predictions until 2029.

(Result) The trend time series reconstruction exhibits a correlation with the reference of up to 0.9. Selecting the parameters with the highest skill levels reveals warming of the North Atlantic sea surface until 2029.

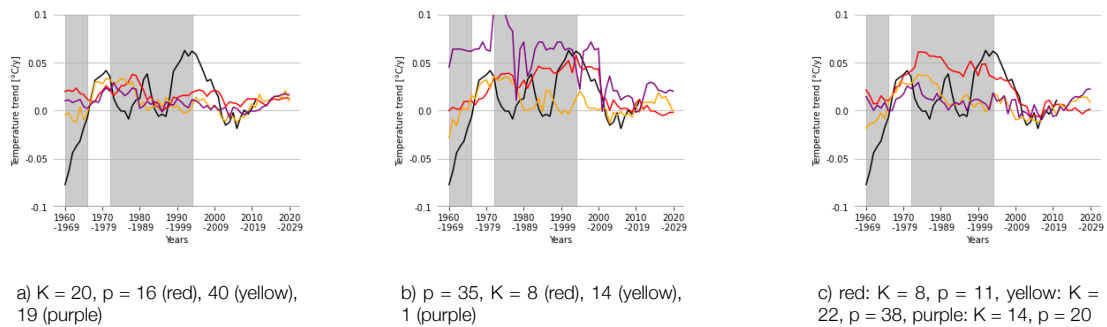


Fig. 11: Trend time series for the best parameters for each sensitivity comparison, c) showing the best parameter set for each similarity definition: 2D RMSE (red), Pearson correlation (yellow) and 3D RMSE (purple). The grey areas are not included in the correlation calculation because they include volcano years.

4. Discussion Outline

Introduction to discussion

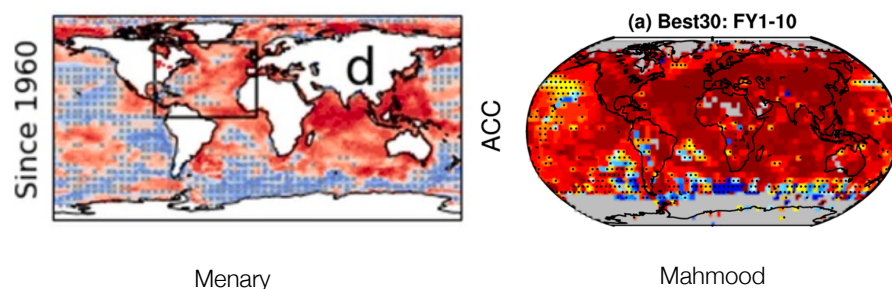
We expanded the analogue method for temperature trend predictions with one simulation ensemble. // We extended the application of the analogue method to predict temperature trends using a single simulation ensemble.

Following a systematic analysis of key unknown parameters, our findings indicate the robustness of the analogue method.

Our study reveals that volcanic eruptions significantly impact the skill of trend predictions. For an accurate estimation of future predictions, where volcanic influence is unknown, it is advisable to exclude this influence.

Literature comparison of the results

(Result) Reconstruction of the trend time series was successful and we find that the analogue methods has improved skill in these regions compared to the persistence/reconstruction made with the reference.



(Result) We can clearly see the influence of the volcanic eruptions. Skill of the reconstruction varies from 0.575 with the inclusion of these years to X without them. Expecting the analogue method to predict external forcings like volcanoes, which it cannot do by design, significantly influences the skill.

(Result) We report that for one simulation ensemble, the MPI-ESM Grand Ensemble, the analogue method is robust. The trend time series reconstruction exhibits a correlation with the reference of up to 0.9.

(Result) Selecting the parameters with the highest skill levels reveals warming of the North Atlantic sea surface until 2029.

Limitations

Exclusion of volcano years: Opens the question of the influence of volcanoes and external forcings in general.

5. Summary and Conclusion

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(*From Intro*) We report that for one simulation ensemble, the MPI-ESM Grand Ensemble, the analogue method is robust. Trend time series reconstruction exhibits a correlation with the reference of up to 0.9, revealing an anticipated warming of the North Atlantic sea surface until 2029.

The results suggest the effectiveness of the analogue method with a single simulation ensemble. A comparison of forecast skill shows that there is not one best parameter set.